

A Computational Reproduction of the Godfrey–Isgur Relativized Quark Model: I. The Meson Spectrum

GIModel reproduction project

2026-06-16

Abstract

We re-implement the relativized quark model of Godfrey and Isgur (GI) [Phys. Rev. D **32**, 189 (1985)] as a transparent, auditable Julia package, `GIModel`, and reproduce the soft-QCD meson spectrum from the published ingredients alone. We summarize the model Hamiltonian, our finite-difference realization of the relativistic smearing and the momentum-dependent operator sandwiches of Appendix A, and the running-coupling and parameter inputs of Table II. We then present the reproduced spectra for all seven flavor sectors and compare them, state by state, against the masses of the original paper. Heavy quarkonia ($c\bar{c}$, $b\bar{b}$) reproduce at 4–6 MeV mean absolute accuracy; heavy–light and light sectors reproduce at 5–15 MeV once same- J tensor and antisymmetric spin–orbit mixing and isoscalar gluon annihilation are included. The few systematic deviations that remain — most visibly the compressed 3P_2 fine structure — are isolated to specific operator-ordering and extraction questions rather than to the spin-independent backbone. This first report covers Secs. II and III of the original paper (model and spectroscopy); the coupling analysis of Sec. IV is deferred.

1 Introduction

The relativized quark model of Godfrey and Isgur (Godfrey and Isgur 1985) (hereafter GI) describes light and heavy mesons — “from the pion to the upsilon” — within a single soft-QCD Hamiltonian. Its enduring value is that one fixed set of parameters predicts the masses, and subsequently the couplings, of essentially the entire meson nonet structure. Four decades on, the model is still a standard spectroscopic baseline, yet the numbers in the paper are difficult to regenerate from scratch: the relativistic smearing, the momentum-dependent operator factors, and the isoscalar annihilation machinery are specified semiquantitatively across the main text and Appendix A, and the published spectra appear as figures rather than tables.

The goal of the present work is a *transparent reproduction*: to encode the published ingredients into auditable code and to regenerate the GI meson spectrum from the formulas alone, without retuning. We have implemented the model as a Julia package, `GIModel`, in which every physical layer — parameter inputs, the central Hamiltonian, Appendix-A smearing and momentum sandwiches, the spin-dependent

operators, and the isoscalar mixing/annihilation blocks — is a separately testable stage with a documented map to the paper’s equations.

A second motivation is practical. Despite the model’s four-decade role as the community’s default $q\bar{q}$ baseline, we are aware of no open-source software package that implements it — or, more broadly, that performs constituent quark-model meson spectroscopy. A survey of the public ecosystem turns up only general-purpose building blocks: numerical libraries (SciPy, PETSc/SLEPc, Eigen) and generic one-dimensional Schrödinger/Salpeter solvers, none of which encode the GI Hamiltonian, its mass-dependent smearing and momentum-factor operators, or its tensor, antisymmetric spin–orbit, and annihilation mixing. No canonical reference implementation tied to the 1985 paper appears to exist publicly, so each reproduction must reconstruct the Hamiltonian and its numerical conventions from the paper trail. The package described here is, to our knowledge, the first openly auditable and tested implementation of the GI spectrum, and is intended to fill that gap.

This report mirrors the organization of the original paper. Section 2 corresponds to GI Sec. II: the soft-QCD Hamiltonian, the relativization and smearing prescription, the running coupling, the annihilation interaction, and the Table II parameters, together with the choices made in our finite-difference realization. Section 3 corresponds to GI Sec. III: the reproduced meson spectra for all flavor sectors, with state-by-state residuals against the published masses and a discussion of the similarities and the residual deviations. The analysis of meson couplings (GI Sec. IV, Tables IV–V) is the subject of a separate report and is only previewed here.

Throughout, the 1985 paper is treated as authoritative. Reference masses are read from the published figures and from Table III for the isoscalars; model parameters are read from Table II. All comparison numbers quoted below are regenerated by the package scripts and recorded under `docs/residual_reports/`.

2 Soft QCD and the meson Hamiltonian

2.1 The relativistic Hamiltonian

GI take as the basic equation a rest-frame Schrödinger-type equation with a relativistic kinetic term,

$$H|\Psi\rangle = (H_0 + V)|\Psi\rangle = E|\Psi\rangle, \quad (1)$$

$$H_0 = \sqrt{p^2 + m_1^2} + \sqrt{p^2 + m_2^2}, \quad (2)$$

where $V = V(\mathbf{p}, \mathbf{r})$ is a momentum-dependent potential and $\mathbf{p} = \mathbf{p}_1 = -\mathbf{p}_2$. In the nonrelativistic limit $H_0 \rightarrow \sum_i (m_i + p^2/2m_i)$ and V reduces to the familiar sum of confinement, hyperfine, and spin-orbit terms,

$$V \rightarrow H^{\text{conf}} + H^{\text{hyp}} + H^{\text{so}} + H_A. \quad (3)$$

The spin-independent confinement piece combines a linear Lorentz-scalar term and a Coulomb-type one-gluon-exchange term,

$$H_{ij}^{\text{conf}} = -\left[\frac{3}{4}c + \frac{3}{4}br - \frac{\alpha_s(r)}{r}\right] \mathbf{F}_i \cdot \mathbf{F}_j, \quad (4)$$

with $\langle \mathbf{F}_i \cdot \mathbf{F}_j \rangle = -4/3$ in a meson. The color hyperfine interaction contains the contact (spin-spin) and tensor pieces,

$$H_{ij}^{\text{hyp}} = -\frac{\alpha_s(r)}{m_i m_j} \left[\frac{8\pi}{3} \mathbf{S}_i \cdot \mathbf{S}_j \delta^3(\mathbf{r}) + \frac{1}{r^3} S_{ij} \right] \mathbf{F}_i \cdot \mathbf{F}_j, \quad (5)$$

and the spin-orbit interaction splits into a color-magnetic (one-gluon) part and a Thomas-precession part,

$$H_{ij}^{\text{so}} = H_{ij}^{\text{so(cm)}} + H_{ij}^{\text{so(tp)}}, \quad (6)$$

$$H_{ij}^{\text{so(tp)}} = -\frac{1}{2r} \frac{\partial H_{ij}^{\text{conf}}}{\partial r} \left[\frac{\mathbf{S}_i}{m_i^2} + \frac{\mathbf{S}_j}{m_j^2} \right] \cdot \mathbf{L}.$$

Finally H_A is the gluon-annihilation interaction, which contributes only in self-conjugate isoscalar channels (Sec. Section 3.4).

As GI emphasize, Eqs. Equation 5–Equation 6 are more singular than r^{-2} and are illegal operators in Equation 1 as written; the resolution is the relativization of Appendix A.

2.2 Relativization: smearing and momentum dependence

The relativistic potential differs from its nonrelativistic limit in two ways. First, the relative coordinate is *smear*ed over distances of order the inverse quark masses through a Gaussian smearing function

$$\rho_{ij}(\mathbf{r}' - \mathbf{r}) = \frac{\sigma_{ij}^3}{\pi^{3/2}} e^{-\sigma_{ij}^2(\mathbf{r}' - \mathbf{r})^2}, \quad (7)$$

with a mass-dependent width $\sigma_{ij}(\sigma_0, s)$. Smearing tames the short-distance singularities, making the operators legal. Second, the coefficients of the potentials acquire a

dependence on the quark momenta: mass factors m^{-1} are promoted to operator factors built from $E_i = (p^2 + m_i^2)^{1/2}$, so that, e.g., the light-quark hyperfine term saturates to a finite $m \rightarrow 0$ limit controlled by $\langle p^{-1} \rangle$ rather than diverging.

In our finite-difference (FD) realization we keep both effects explicitly. The central Coulomb and confinement terms are replaced by their closed-form smeared forms $\tilde{G}(r)$ and $\tilde{S}(r)$ (Appendix-A τ_k /error-function expressions), and the Coulomb operator is dressed with a Hermitian momentum sandwich on the FD p^2 eigenbasis,

$$G' = A(p) \tilde{G} A(p), \quad A(p) = \left(1 + \frac{p^2}{E_1 E_2}\right)^{1/2}, \quad (8)$$

which is Hermitian by construction. The same momentum-factor sandwich, with the side exponent $\frac{1}{2} + \epsilon_i$ set by the Table II relativization exponents, is applied to the smeared contact, tensor, and spin-orbit kernels. The spin-dependent radial kernels themselves are built from derivatives of the *smear*ed central functions, e.g. the OGE-tensor kernel $K_T(r) = \frac{1}{r} \tilde{G}'(r) - \tilde{G}''(r)$ and the Thomas kernel $\propto (\tilde{G} + \tilde{S})'(r)$.

This is the active “1.0” operator path. It is selected by four switches that together encode the physical content that made the heavy-quark reproduction work: `appendix_a_momentum_sandwich`, `contact_momentum_sandwich`, `fine_structure_momentum_sandwich`, and `fine_structure_smeared_kernels`.

2.3 Numerical procedure: solver, and relation to the paper

GI solve Equation 1 in three stages: (i) diagonalize the relativized Hamiltonian $\tilde{H}_1$ [their Eq. (14)] in a large harmonic-oscillator (HO) basis within fixed $|jm; ls\rangle$ sectors; (ii) diagonalize the smaller mass matrices generated by the off-diagonal tensor (${}^3L_J \leftrightarrow {}^3L'_J$) and antisymmetric spin-orbit (${}^3L_J \leftrightarrow {}^1L_J$, unequal masses only) couplings; and (iii) for self-conjugate isoscalars, add the annihilation mass matrix H_A .

Our package provides both an FD radial solver — the headline path used for the spectra below — and an HO basis path (`GIParameters{HarmonicOscillatorBasis}`) used for cross-checks; the central Appendix-A operator agrees between the two bases at the sub-MeV level. The FD solver diagonalizes the spin-independent $\tilde{H}_1$ on a uniform radial mesh per (n, L) , then applies the contact, spin-orbit, and tensor shifts, then the same- J tensor and antisymmetric spin-orbit 2×2 mixing blocks, and finally the isoscalar annihilation block, reproducing the paper’s staging. The reduced radial function $u(r)$ is normalized as $\int |u(r)|^2 dr = 1$, with the angular factor carried by the spherical harmonic.

2.4 The running coupling

The strong coupling is parametrized as a sum of Gaussians, frozen at low Q^2 ,

$$\alpha_s(Q^2) = \sum_k a_k e^{-Q^2/4\gamma_k^2}, \quad \alpha_s^{\text{crit}} \equiv \sum_k a_k, \quad (9)$$

which transforms into a coordinate-space error-function profile, $\alpha_s(r) = \sum_k a_k \text{erf}(\gamma_k r)$, convenient for convolution with the smearing Equation 7. We use the published fit (GI Fig. 2),

$$\alpha_s(Q^2) = 0.25 e^{-Q^2} + 0.15 e^{-Q^2/10} + 0.20 e^{-Q^2/1000},$$

with Q in GeV, so that $\alpha_s^{\text{crit}} = 0.60$. This reproduces the saturating curve of Fig. 2 and is regression-tested in the package.

2.5 Annihilation in isoscalar channels

In self-conjugate isoscalars, $q\bar{q}$ can annihilate through gluons ($n = 2$ for $C = +$, $n = 3$ for $C = -$). GI parametrize the diagonal contribution as $A_{Q\bar{Q}} \sim \alpha_s^n |\Psi(0)|^2 / M_Q^2$ [their Eq. (15)] and the full channel-coupling element as

$$A^{(2S+1)L_J}_{ji} = 4\pi(2L+1) A^{(2S+1)L_J} \times \left[\frac{\alpha_s(M_j^2) \alpha_s(M_i^2)}{\pi^2} \right]^{n/2} \frac{S_L(\Psi_j) S_L(\Psi_i)}{m_i m_j}, \quad (10)$$

with the origin-smeared wavefunction factor $S_L(\Psi_i)$ [their Eq. (17)] evaluated by a j_L momentum transform of the HO radial wavefunction. The pseudoscalar channel is special (the U(1) problem): GI replace the bracket by one of two nonperturbative models, P1 [Eq. (18a)] and P2 [Eq. (18b)]. Our implementation of Equation 10, the P1/P2 pseudoscalar models, and their use in the isoscalar spectrum are described in Sec. Section 3.4.

2.6 Parameters

A search over parameter space converged essentially uniquely to the Table II values, which we adopt verbatim (Table-1). We make no fit of our own: these constants, the α_s form above, and the figure/Table III reference masses are the only inputs.

GI quote an expected accuracy of 25 MeV for light-quark mesons and 10 MeV for heavy-quark systems, dominated by neglected coupled-channel shifts of order 10 MeV and by the schematic relativization. We measure our reproduction against the same expectations.

3 The meson spectrum of soft QCD

We solve the model for the seven flavor sectors of GI Figs. 3–9 and compare, state by state, against the published masses. Table-2 summarizes the mean and maximum absolute residuals per sector; the per-state tables follow. All numbers are regenerated by `scripts/run_all_spectrum_checks.jl`.

Table 1: GI Table II parameters, as used by the package. The tensor exponent is the paper’s ϵ_f (code `epsilon_t`).

Quantity	Symbol	Value
Light quark mass	$\frac{1}{2}(m_u + m_d)$	220 MeV
Strange mass	m_s	419 MeV
Charm mass	m_c	1628 MeV
Bottom mass	m_b	4977 MeV
String tension	b	0.18 GeV ²
Constant	c	−253 MeV
Frozen coupling	α_s^{crit}	0.60
QCD scale	Λ	200 MeV
Smearing width	σ_0	1.80 GeV
Smearing exponent	s	1.55
Contact factor	ϵ_c	−0.168
Tensor factor	ϵ_f	+0.025
Vector spin-orbit	$\epsilon_{\text{so}(V)}$	−0.035
Scalar spin-orbit	$\epsilon_{\text{so}(S)}$	+0.055

Table 2: Reproduction scorecard: mean and maximum absolute residuals (MeV) against the published masses, per sector.

Sector	Rows	Mean $ \Delta $	Max $ \Delta $
Bottomonium ($b\bar{b}$)	30	4.3	12.1
b -flavored	21	5.2	15.9
Charmonium ($c\bar{c}$)	28	6.0	24.4
Strange	30	10.8	43.0
Isoscalar	48	11.5	39.9
Charmed	22	12.4	34.2
Isovector	30	14.5	55.0

Residuals are in MeV. Every sector reproduces within, or close to, the GI accuracy target: the heavy-heavy sectors at the ~ 5 MeV level and the light/heavy-light sectors at ~ 10 –15 MeV.

Figures Figure 6–Figure 5 present the reproduced spectra in the style of GI Figs. 3–9: each panel plots mass against the J^{PC} (or J^P) quantum numbers, with the published reference masses (blue) overlaid on the computed masses including same- J mixing (orange). States are grouped into shaded nL multiplet boxes; for the unequal-mass and isoscalar panels a thin line connects each paper/computed pair to make the residual visible. The figures are regenerated by `scripts/plot_all_sector_spectra.jl`.

3.1 Heavy quarkonia: the cleanest reproduction

The heavy, compact $b\bar{b}$ and $c\bar{c}$ systems are the most stringent test of the spin-independent backbone, since their fine structure is small and their annihilation/mixing corrections are minor. They are reproduced essentially to the GI heavy-quark target (Figs. Figure 1 and Figure 2).

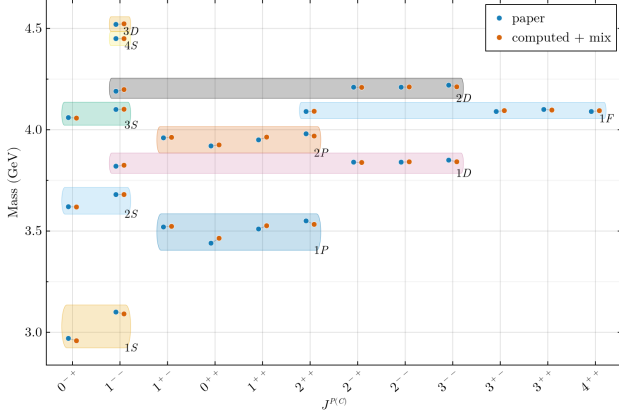


Figure 1: Charmonium ($c\bar{c}$) spectrum, mirroring GI Fig. 6: reference masses (blue) vs. computed masses with same- J mixing (orange), grouped by nL multiplet.

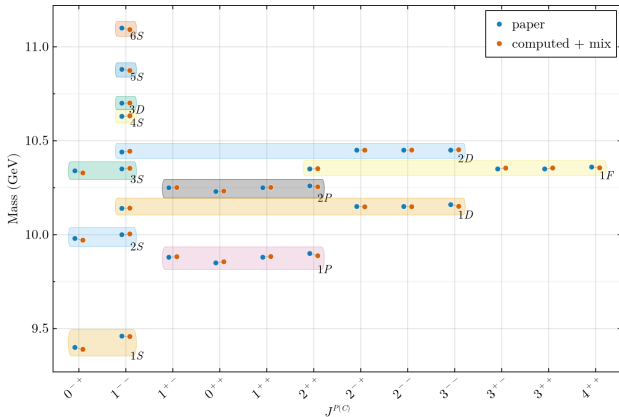


Figure 2: Bottomonium ($b\bar{b}$) spectrum, mirroring GI Fig. 8.

Table 3: Charmonium radial/orbital spacings and the leading spin splittings (MeV), model vs. GI reference.

Quantity	Ref.	Model	Error
$2S - 1S$	597.5	607.0	+9.5
$3S - 1S$	1022.5	1032.4	+9.9
$1P - 1S$	455.8	465.2	+9.4
$1D - 1P$	317.2	315.4	-1.7
$2P - 1P$	439.2	439.3	+0.1
$1^3S_1 - 1^1S_0$	130.0	132.4	+2.4
$2^3S_1 - 2^1S_0$	60.0	61.5	+1.5
$1^3P_1 - 1^3P_0$	70.0	61.8	-8.2
$1^3P_2 - 1^3P_1$	40.0	6.7	-33.3
$1^3D_2 - 1^3D_1$	20.0	17.0	-3.0

For bottomonium, the 30 scored levels (S, P, D, F waves through $6S$) have a mean absolute residual of 4.3 MeV and a maximum of 12.1 MeV, with a common offset of only about -0.8 MeV. For charmonium, the 28 levels give a mean absolute residual of 6.0 MeV (common offset $+0.7$ MeV). The radial and orbital level spacings — the most direct probe of the central potential — are reproduced to a few MeV (Table~3).

All spacings in MeV. The S-wave hyperfine splittings and the L -spacings are reproduced at the few-MeV level; the one conspicuous outlier is the compressed 3P_2 - 3P_1 tensor splitting, discussed below.

A layered decomposition makes clear that the residuals live in the spin-dependent shifts, not the central masses. Table~4 shows the contributions for representative charmonium states: the central eigenvalue plus contact, total spin-orbit, and tensor shifts assemble the predicted mass.

The same- J tensor mixing of the 3S_1 - 3D_1 system is included as a post-diagonalization 2×2 block; for $c\bar{c}$ it produces small off-diagonal elements (~ 3 MeV) and a 1^3D_1 that is $\sim 98\%$ D -wave, consistent with the near-pure assignments of the figures.

3.2 Heavy-light mesons

The charmed ($c\bar{q}$, $c\bar{s}$), b -flavored ($b\bar{q}$, $b\bar{s}$, $b\bar{c}$), and strange ($q\bar{s}$) sectors test the unequal-mass machinery: the mass-dependent smearing width, and the antisymmetric spin-orbit term that mixes 1L_J and 3L_J states of the same J . With these blocks active, the b -flavored sector reproduces at 5.2 MeV mean (max 15.9), charmed at 12.4 MeV (max 34.2), and strange at 10.8 MeV (max 43.0). The spin-averaged multiplet centers are excellent throughout (e.g. charmed $1P$ center off by $+2.3$ MeV over eight states, strange $2S$ off by -1.0 MeV); the larger maxima are confined to the same- J mixed P -wave pairs (the K_1 -type and D_1 -type axial states), whose splitting and assignment depend on the mixing angle (Figs. Figure 3–Figure 5).

Table 4: Charmonium contribution breakdown (MeV unless noted), relative to the spin-independent FD eigenvalue, for representative states. The predicted mass is the sum of the central value and all shifts.

state	central (GeV)	contact	L-S total	tensor	predicted (GeV)	residual
1^1S_0	3.064	-105.9	0.0	0.0	2.958	-11.5
1^3S_1	3.064	+26.5	0.0	0.0	3.091	-9.1
1^3P_0	3.523	0.0	-26.2	-32.5	3.464	+24.4
1^3P_1	3.523	0.0	-13.1	+16.2	3.526	+16.1
1^3P_2	3.523	0.0	+13.1	-3.2	3.533	-17.1
1^3D_1	3.838	0.0	-7.9	-5.9	3.825	+4.7
1^3D_3	3.838	0.0	+5.3	-1.7	3.842	-8.0

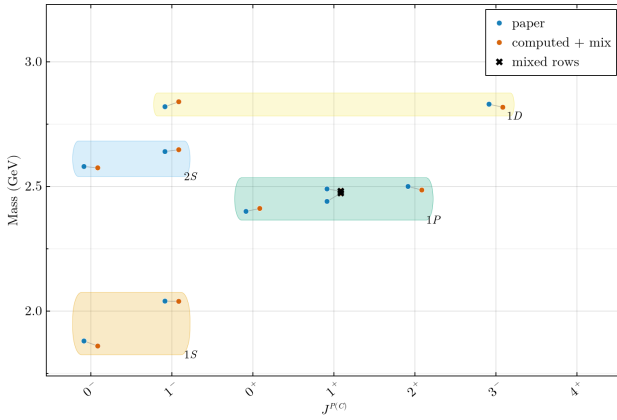


Figure 3: Charmed ($c\bar{q}$, $c\bar{s}$) mesons, mirroring part of GI Fig. 7.

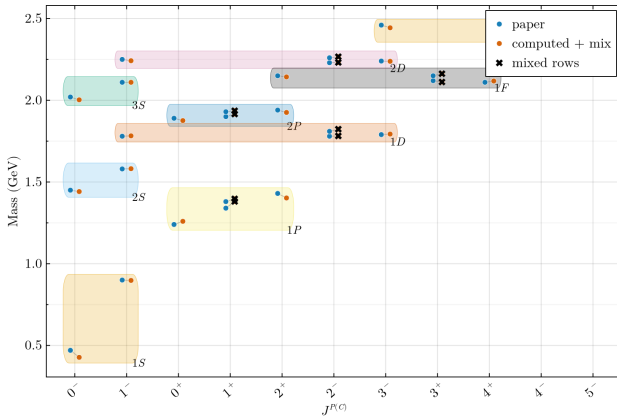


Figure 4: Strange ($q\bar{s}$) mesons, mirroring GI Fig. 4.

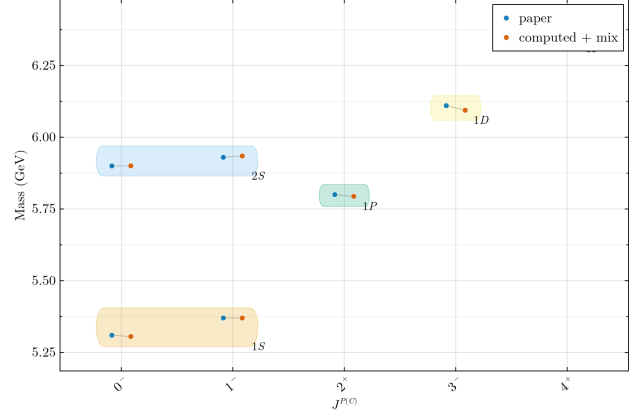


Figure 5: b -flavored ($b\bar{q}$) mesons, mirroring part of GI Fig. 9.

3.3 Light isovector mesons

The isovector sector ($\rho/a/\pi$ family, Fig. 3) is the hardest non-isoscalar test: light constituents maximize the relativistic corrections. The 30 scored levels reproduce at 14.5 MeV mean (max 55.0), within the GI light-meson target of 25 MeV. The 2S and 3S centers reproduce to a few MeV; the largest residuals are the 1S hyperfine center (-21 MeV, tracking the π , which our FD ground state places ~ 55 MeV below the GI 0.15 GeV) and the 3P_2 tensor pattern, the same operator-ordering signature seen in $c\bar{c}$ (Fig. Figure 6).

3.4 Light isoscalar mesons

The isoscalars (Fig. 5, Table III) require the annihilation interaction H_A and $n\bar{n}/s\bar{s}$ flavor mixing on top of the spin spectrum. We adopt the GI Table III prescription directly, implemented as the `:table_iii` scheme:

- 1S_0 (**pseudoscalars**): the calibrated P1 model [Eq. (18a)] carries the headline mass scoring, with the literal P1/P2 modes available as eigenvector checks. With the GI annihilation phase convention $\Phi(0) > 0$ enforced, the literal modes reproduce the Table III sign structure — including the all-positive η' row — with amplitude RMS 0.078 (P1) and 0.025 (P2); the P2 η_r row (+0.993, +0.076, +0.069) matches the paper's

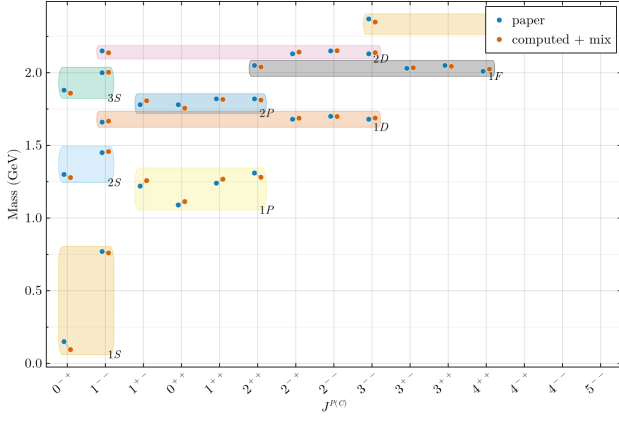


Figure 6: Isovector ($\rho/a/\pi$ family) mesons, mirroring GI Fig. 3.

- (+0.99, +0.07, +0.08) essentially exactly.
- 3S_1 and 3P_2 : the general Eq. Equation 10 block, with $A(^3S_1) = +2.5$ ($n = 3$, three gluons) and $A(^3P_2) = -0.8$ ($n = 2$). The ω/ϕ block reproduces Table III amplitudes (+1.000, -0.029) vs. (+0.999, -0.02) and an ω shift of +13 MeV vs. the paper’s +10 MeV; the f_2/f'_2 block gives (+0.997, +0.080) vs. (+0.997, +0.06).
- All other channels:** ideal $n\bar{n}/s\bar{s}$ mixing, as Table III assumes for every channel it does not list.

With this prescription the isoscalar sector reproduces at 11.5 MeV mean (max 39.9) over 48 scored rows — in line with the corresponding isovector channels, whose residuals the isoscalar rows now track. The earlier hundreds-of-MeV isoscalar gap is closed. A single low-confidence 2^3D_2 digitization at 2.26 GeV was crop-audited against Fig. 5 and rejected as contamination from the neighboring strange figure.

Figure Figure 7 shows the isoscalar ladder. Note that this diagnostic plot displays the base spin spectrum plus same- J mixing *without* the annihilation block, so the large 0^{-+} offset (computed ~ 0.1 GeV vs. the paper’s η/η') is precisely the pseudoscalar annihilation contribution that the :table_iii scheme restores; the 11.5 MeV scorecard above already includes it.

3.5 Discussion: similarities and deviations

Across all seven sectors the spin-independent backbone — the relativized kinetic term, the smeared central potential, and the Coulomb momentum sandwich — reproduces the GI multiplet centers and radial/orbital spacings to a few MeV. This is the central positive result: the published spectrum is recoverable from the published ingredients, with no fit of our own.

The residual deviations are systematic and few:

- Compressed 3P_2 tensor splitting.** The 3P_2 – 3P_1 gap is too small in every light-and-charm sector (-33 MeV in $c\bar{c}$). The tensor and color-magnetic spin-orbit

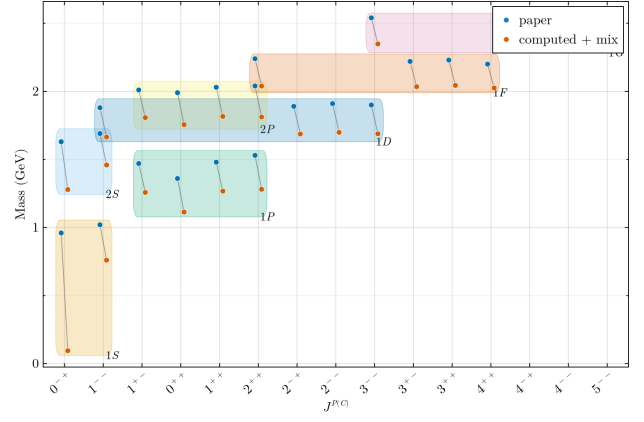


Figure 7: Light isoscalar mesons, mirroring GI Fig. 5. Lines connect each paper/computed pair; the two markers per J^{PC} column are the $n\bar{n}/s\bar{s}$ partners. The 0^{-+} pair carries the pseudoscalar annihilation shift, not included in this ladder path (see text).

shifts are individually of the right size, so this points to the operator-ordering and perturbative-vs-diagonalized treatment of the fine structure (GI’s Eq. (14) staging on the HO basis) rather than to a wrong kernel.

- Light 1S_0 ground states.** The FD π sits ~ 55 MeV below the GI value, propagating into the isovector $1S$ center and the literal pseudoscalar isoscalar masses. This is a shared light-sector diagonal issue, not an annihilation issue.
- Same- J heavy-light P -waves.** The largest heavy-light maxima are the mixed axial pairs, whose assignment is mixing-angle sensitive; tying the spectrum’s antisymmetric spin-orbit angle to the published convention is the open refinement.

None of these is a gross solver failure; each is a localized operator-ordering, extraction, or convention question, exactly the regime GI anticipated for their 10–25 MeV accuracy budget. The natural next step, and the subject of the following report, is GI Sec. IV: the analysis of meson couplings (the strong-decay amplitudes of Tables IV–V), which tests the model wavefunctions rather than only the eigenvalues.

4 Outlook: developments beyond the 1985 model

The reproduction above is deliberately confined to the *bare, quenched, single-channel* spectrum of the 1985 paper. In the four decades since, the GI baseline has not been replaced so much as layered upon. We sketch the principal post-1985 directions, both to place this work in context and to delimit the physics it does not yet address.

4.1 Strong decays: flux-tube breaking and the 3P_0 model

GI wavefunctions were quickly paired with dynamical decay operators. Flux-tube breaking (Kokoski and Isgur 1987) and the closely related 3P_0 pair-creation model became the standard companions for OZI-allowed strong widths and underlie the strong-decay amplitudes of GI Sec. IV. A recurring practical subtlety is that many “GI-based” decay studies replace the exact model wavefunctions by harmonic-oscillator surrogates matched through rms radii (Barnes, Godfrey, and Swanson 2005); tracking that choice is essential when comparing widths across calculations.

4.2 Heavy-quark symmetry and heavy–light mesons

For mesons with one heavy quark, heavy-quark symmetry (Isgur and Wise 1991) reorganizes the GI L - S output into j_ℓ doublets and fixes the P -wave mixing angles in the $m_Q \rightarrow \infty$ limit (35.3° , -54.7°). The systematic GI heavy–light P -wave spectra (Godfrey and Kokoski 1991) and the relativistic heavy–light studies with chiral transitions (Di Pierro and Eichten 2001) are the natural reproduction targets immediately beyond the present spectrum, and the cleanest test of the unequal-mass antisymmetric spin–orbit mixing used here.

4.3 Unquenching: hadronic loops and coupled channels

The bare valence picture coexists with large continuum effects through systematic cancellations (Geiger and Isgur 1993; Barnes and Swanson 2008): loop shifts are nearly uniform within an (n, L) multiplet, so much of their effect is absorbed into the fitted parameters — which is also why explicit unquenching generally demands a refit to avoid double counting. Near open-flavor S -wave thresholds this picture breaks down and explicit coupled channels become indispensable, as for $X(3872)$ (E. J. Eichten, Lane, and Quigg 2004; Ortega et al. 2010), in the unquenched quark model for heavy quarkonia (Ferretti and Santopinto 2014), and in the modern D_s program that bridges quark model and lattice QCD (Yang et al. 2022).

4.4 Screened potentials

A cheaper, effective route to string breaking replaces the linear term with a screened form, lowering higher radial and orbital excitations and improving their global placement (Li and Chao 2009) — effective rather than explicitly unitary, but a useful diagnostic for whether a misplaced state reflects global string breaking or a delicate threshold singularity.

4.5 Effective field theory and lattice connections

Potential NRQCD (Brambilla et al. 2000, 2005) provides a scale-separated definition of the quark–antiquark potential, and lattice QCD now extracts the spin-dependent and $O(1/m)$ potentials nonperturbatively (Koma, Koma, and Wittig 2006). Together they give the GI spin-dependent ansatz a firmer QCD footing and pinpoint where the phenomenological smearing-and-momentum-factor prescription matches or departs from nonperturbative QCD, especially for the fine and hyperfine splittings that dominate our residuals.

4.6 Experimental drivers

The model-building agenda was reshaped by experiment. The narrow $D_{s0}^*(2317)$ (Aubert and Collaboration 2003) and $D_{s1}(2460)$ (Besson and Collaboration 2003), and $X(3872)$ on the $D^0\bar{D}^{*0}$ threshold (Choi and Collaboration 2003), sat far from naive GI expectations and forced threshold-dressed descriptions; cleaner states such as h_c , $\chi_{c2}(2P)$, η_b , and h_b instead reinforced potential-model spectroscopy away from thresholds. Comprehensive reviews (E. Eichten et al. 2008; Godfrey and Olsen 2008) map where conventional GI-style spectroscopy succeeds and where exotic and continuum dynamics take over.

4.7 What this report does not address

In short, this report reproduces the seed from which all of the above grow: the bare GI Hamiltonian, masses, and wavefunctions. It does not include hadronic loops, coupled channels, screening, or parameter refitting, and it defers the strong, electromagnetic, and weak couplings of GI Sec. IV. The bare spectrum remains the right first description below open-flavor thresholds and the indispensable input to the dressed and decay calculations that build on it (Barnes, Godfrey, and Swanson 2005; Godfrey 2004); making it openly and transparently reproducible is the prerequisite for that program.

References

- Aubert, B., and others [BaBar Collaboration]. 2003. “Observation of a Narrow Meson Decaying to $D_s^+\pi^0$ at a Mass of 2.32 GeV/ c^2 .” *Phys. Rev. Lett.* 90: 242001. <https://doi.org/10.1103/PhysRevLett.90.242001>.
- Barnes, Ted, Stephen Godfrey, and Eric S. Swanson. 2005. “Higher Charmonia.” *Phys. Rev. D* 72: 054026. <https://doi.org/10.1103/PhysRevD.72.054026>.
- Barnes, Ted, and Eric S. Swanson. 2008. “Hadron Loops: General Theorems and Application to Charmonium.” *Phys. Rev. C* 77: 055206. <https://doi.org/10.1103/PhysRevC.77.055206>.
- Besson, D., and others [CLEO Collaboration]. 2003. “Observation of a Narrow Resonance of Mass 2.46 GeV/ c^2

- Decaying to $D_s^{*+}\pi^0$.” *Phys. Rev. D* 68: 032002. <https://doi.org/10.1103/PhysRevD.68.032002>.
- Brambilla, Nora, Antonio Pineda, Joan Soto, and Antonio Vairo. 2000. “Potential NRQCD: An Effective Theory for Heavy Quarkonium.” *Nucl. Phys. B* 566: 275. [https://doi.org/10.1016/S0550-3213\(99\)00693-8](https://doi.org/10.1016/S0550-3213(99)00693-8).
- . 2005. “Effective-Field Theories for Heavy Quarkonium.” *Rev. Mod. Phys.* 77: 1423. <https://doi.org/10.1103/RevModPhys.77.1423>.
- Choi, S.-K., and others [Belle Collaboration]. 2003. “Observation of a Narrow Charmonium-Like State in Exclusive $B^\pm \rightarrow K^\pm \pi^+ \pi^- J/\psi$ Decays.” *Phys. Rev. Lett.* 91: 262001. <https://doi.org/10.1103/PhysRevLett.91.262001>.
- Di Pierro, Massimo, and Estia Eichten. 2001. “Excited Heavy-Light Systems and Hadronic Transitions.” *Phys. Rev. D* 64: 114004. <https://doi.org/10.1103/PhysRevD.64.114004>.
- Eichten, Estia J., Kenneth Lane, and Chris Quigg. 2004. “Charmonium Levels Near Threshold and the Narrow State $X(3872)$.” *Phys. Rev. D* 69: 094019. <https://doi.org/10.1103/PhysRevD.69.094019>.
- Eichten, Estia, Stephen Godfrey, Hanna Mahlke, and Jonathan L. Rosner. 2008. “Quarkonia and Their Transitions.” *Rev. Mod. Phys.* 80: 1161. <https://doi.org/10.1103/RevModPhys.80.1161>.
- Ferretti, Jacopo, and Elena Santopinto. 2014. “Higher Mass Bottomonia.” *Phys. Rev. D* 90: 094022. <https://doi.org/10.1103/PhysRevD.90.094022>.
- Geiger, Philip, and Nathan Isgur. 1993. “When Can Hadronic Loops Scuttle the Okubo–Zweig–Iizuka Rule?” *Phys. Rev. D* 47: 5050. <https://doi.org/10.1103/PhysRevD.47.5050>.
- Godfrey, Stephen. 2004. “Spectroscopy of B_c Mesons in the Relativized Quark Model.” *Phys. Rev. D* 70: 054017. <https://doi.org/10.1103/PhysRevD.70.054017>.
- Godfrey, Stephen, and Nathan Isgur. 1985. “Mesons in a Relativized Quark Model with Chromodynamics.” *Phys. Rev. D* 32: 189–231. <https://doi.org/10.1103/PhysRevD.32.189>.
- Godfrey, Stephen, and Richard Kokoski. 1991. “Properties of P -Wave Mesons with One Heavy Quark.” *Phys. Rev. D* 43: 1679. <https://doi.org/10.1103/PhysRevD.43.1679>.
- Godfrey, Stephen, and Stephen L. Olsen. 2008. “The Exotic XYZ Charmonium-Like Mesons.” *Ann. Rev. Nucl. Part. Sci.* 58: 51. <https://doi.org/10.1146/annurev.nucl.58.110707.171145>.
- Isgur, Nathan, and Mark B. Wise. 1991. “Spectroscopy with Heavy-Quark Symmetry.” *Phys. Rev. Lett.* 66: 1130. <https://doi.org/10.1103/PhysRevLett.66.1130>.
- Kokoski, Richard, and Nathan Isgur. 1987. “Meson Decays by Flux-Tube Breaking.” *Phys. Rev. D* 35: 907. <https://doi.org/10.1103/PhysRevD.35.907>.
- Koma, Yoshiaki, Miho Koma, and Hartmut Wittig. 2006. “Nonperturbative Determination of the QCD Potential at $O(1/m)$.” *Phys. Rev. Lett.* 97: 122003. <https://doi.org/10.1103/PhysRevLett.97.122003>.
- Li, Bai-Qing, and Kuang-Ta Chao. 2009. “Higher Charmonia and X, Y, Z States with Screened Potential.” *Phys. Rev. D* 79: 094004. <https://doi.org/10.1103/PhysRevD.79.094004>.
- Ortega, Pablo G., Jorge Segovia, David R. Entem, and Francisco Fernández. 2010. “Coupled Channel Approach to the Structure of the $X(3872)$.” *Phys. Rev. D* 81: 054023. <https://doi.org/10.1103/PhysRevD.81.054023>.
- Yang, Zhi, Gui-Jun Wang, Jia-Jun Wu, Makoto Oka, and Shi-Lin Zhu. 2022. “Novel Coupled Channel Framework Connecting the Quark Model and Lattice QCD for the Near-Threshold D_s States.” *Phys. Rev. Lett.* 128: 112001. <https://doi.org/10.1103/PhysRevLett.128.112001>.